6 **(a)** State the principle of superposition.

.....

..... [1]

(b) A 15.0 m long glass tube has a loudspeaker, connected to a signal generator, placed at one end.

A microphone connected to a cathode-ray oscilloscope (CRO) can be moved through the tube as shown in Fig. 6.1.

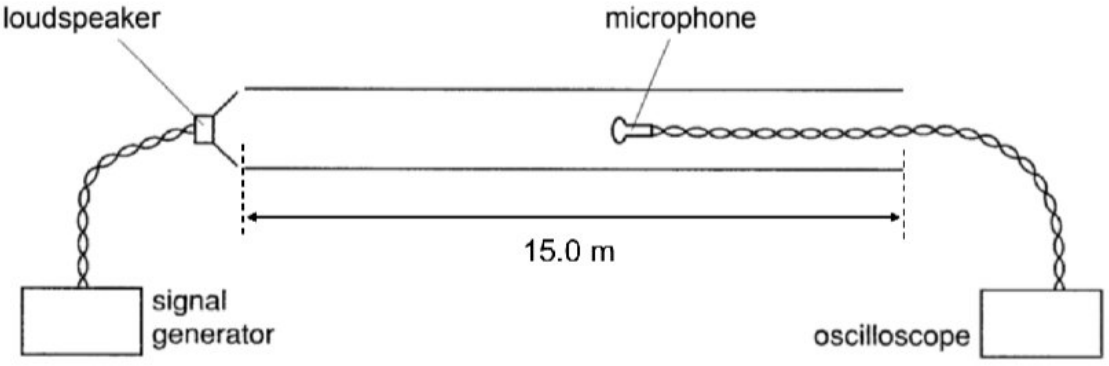


Fig. 6.1

State the conditions for the formation of a stationary wave and explain how an observable stationary wave is formed in the long glass tube.

.....

.....

.....

.....

.....

..... [3]

(c) The speaker produces sound waves with speed 320 m s^{-1} and an amplitude of 0.060 mm.

The frequency of the wave is adjusted until a loud sound is heard.

(i) State the maximum amplitude of the stationary wave

maximum amplitude = ... mm [1]

(ii) The microphone is moved from one end to the other within the 15.0 m long glass tube and detects a total of three pressure antinodes.

Complete Fig. 6.2 to represent the stationary wave in the glass tube.

Fig. 6.2

[2]

(iii) Using your answer in **(c)(ii)**, determine the frequency of the stationary wave.

frequency = ... Hz [1]

[Total: 8]